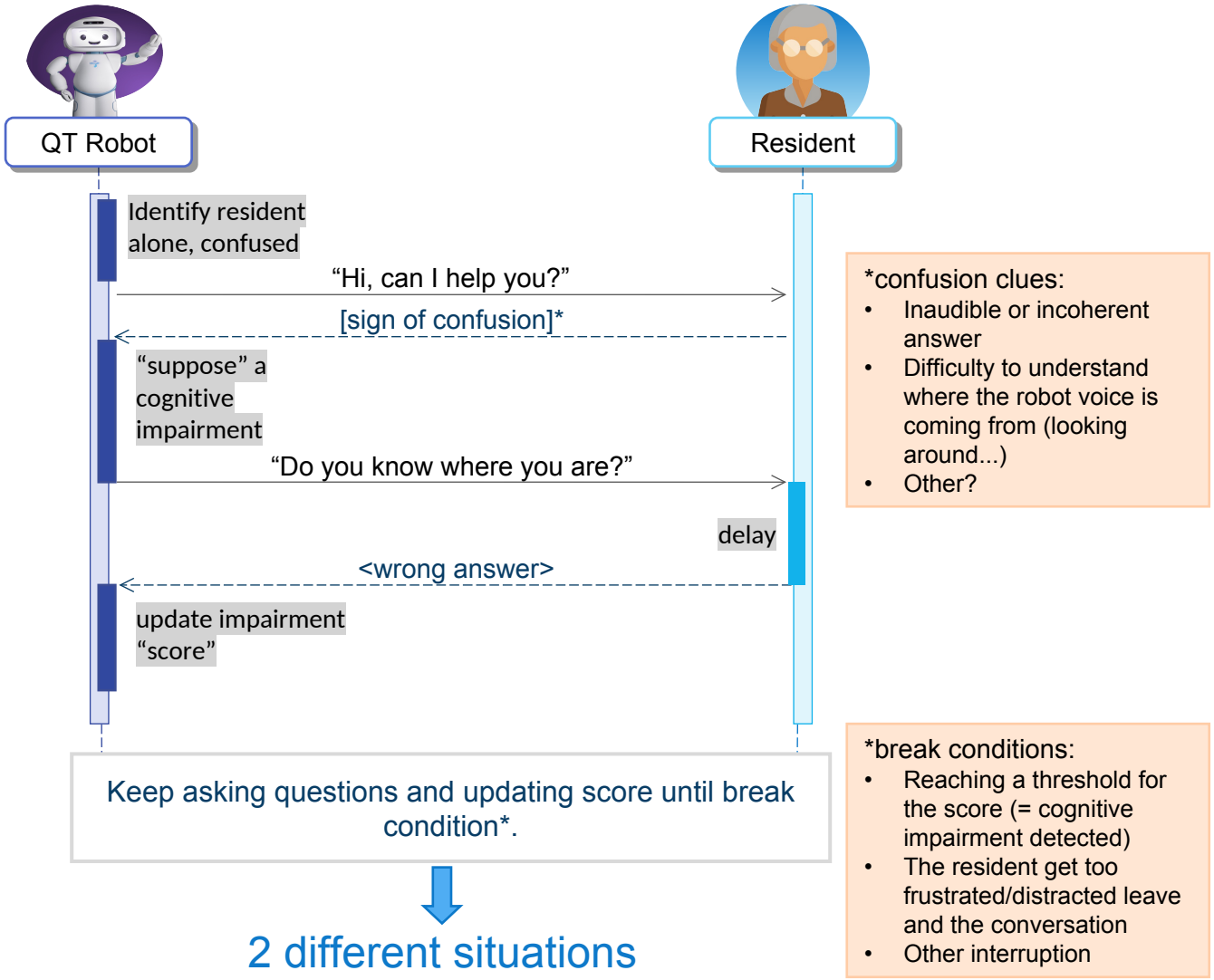
Scenario for human-robot interactions

Member of a nursing home care-team

One-to-one interaction

Context : The robot sees a resident alone and starts an interaction

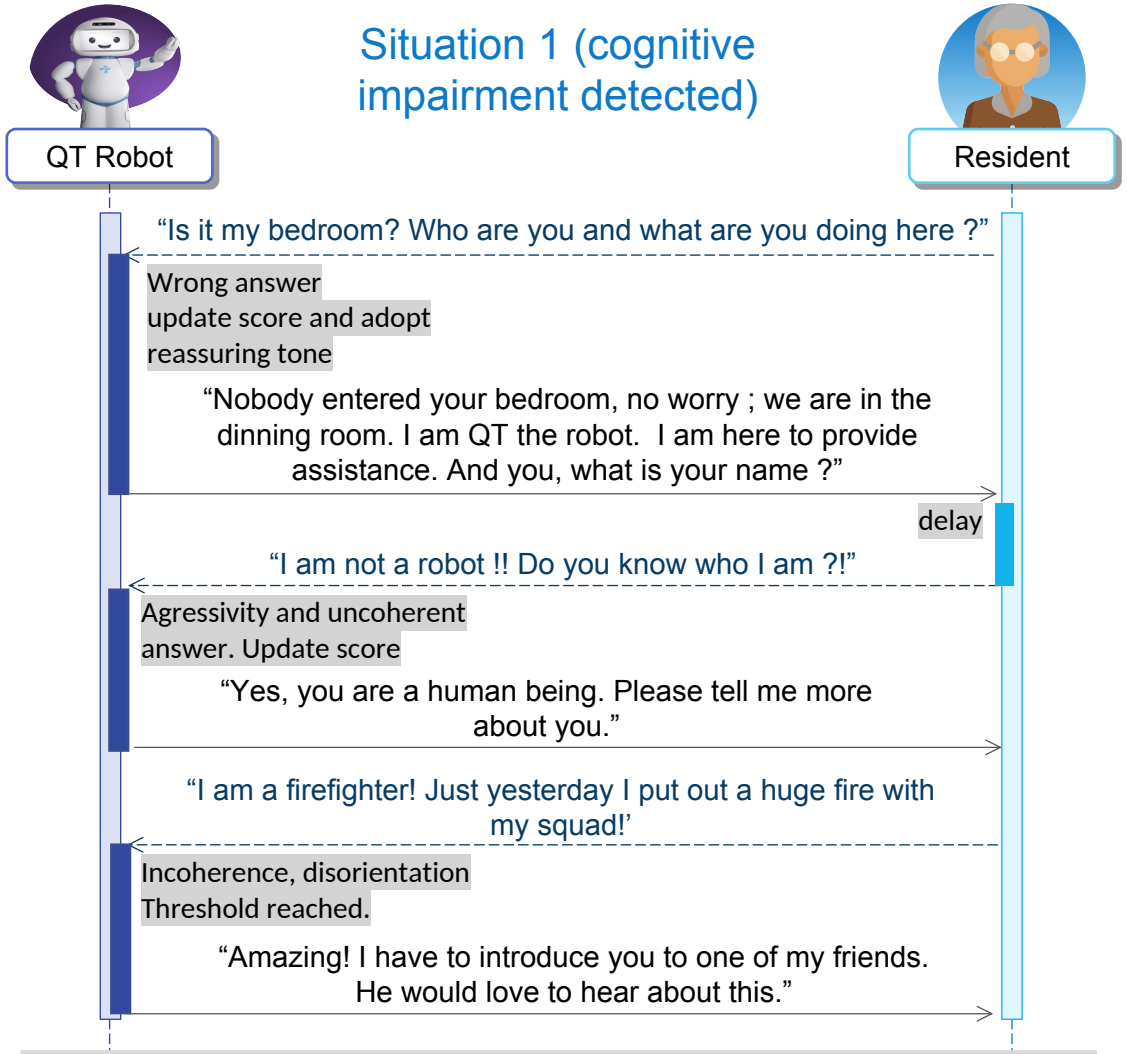


- Basic things to ask :
  - Ask about the location
  - Ask about the date
  - Ask about what their last meal
  - Ask about the family
  - Ask about the actuality (who is the French president?)

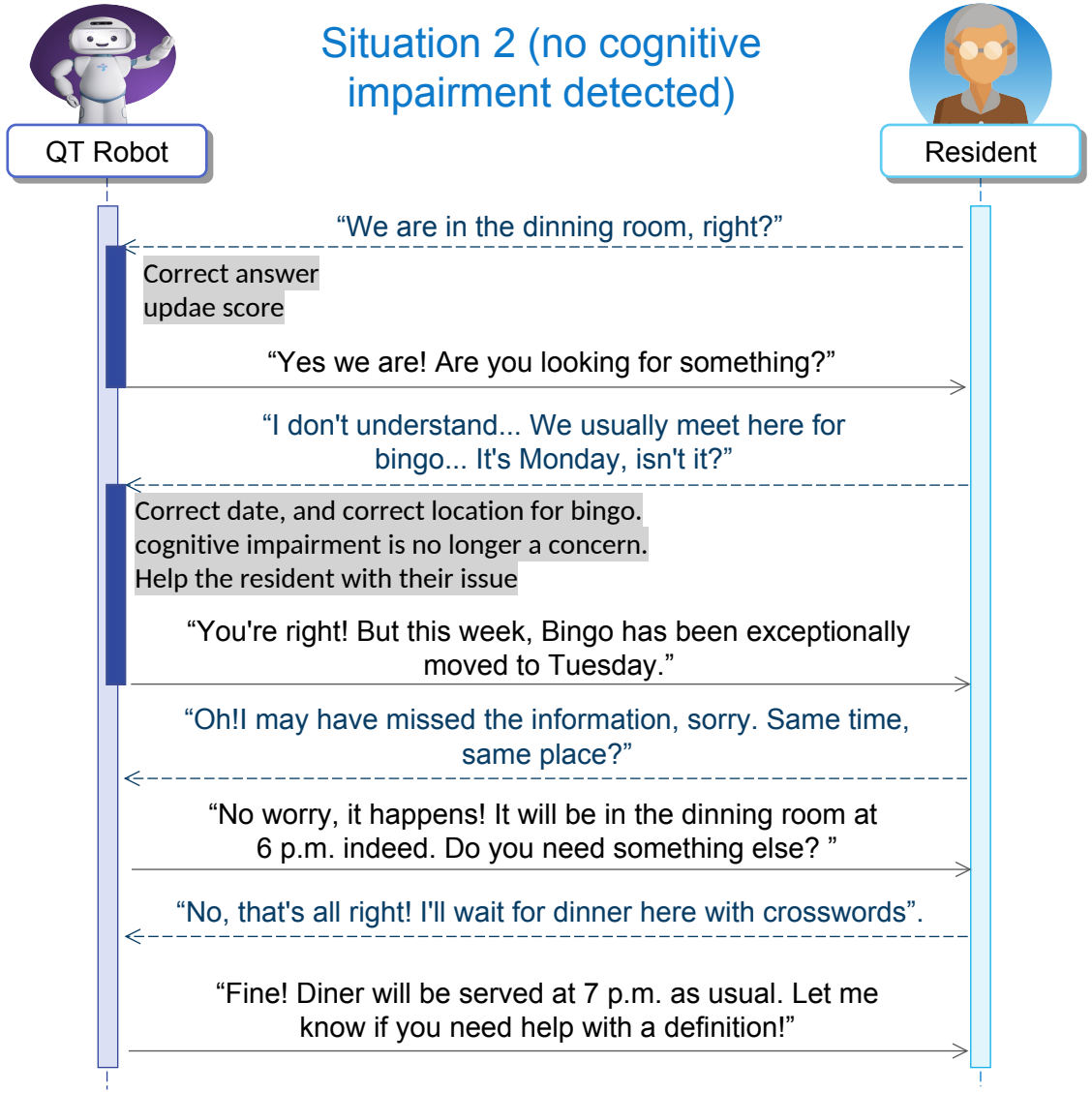
- Detect frustration from the resident:
  - Detect intense emotions (anger)
  - Detect negative answers ("I don't know", systematic 'no')

One-to-one interaction (following)

Context : The robot just asked the resident where he is after suspecting a cognitive impairment

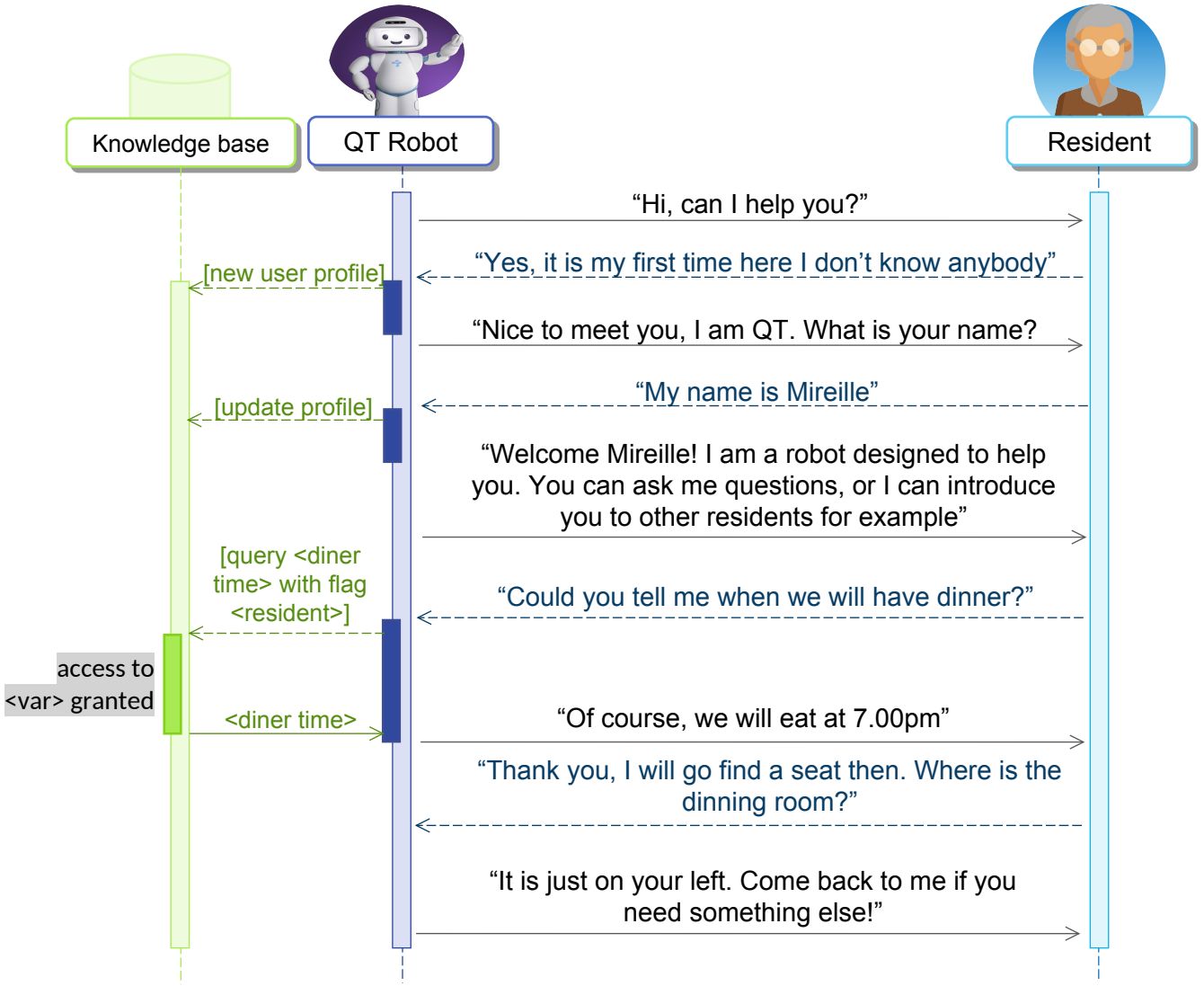


Bring the resident to a care team member OR notify a Care Team member to come and provide assistance for a disoriented resident



One-to-one interaction

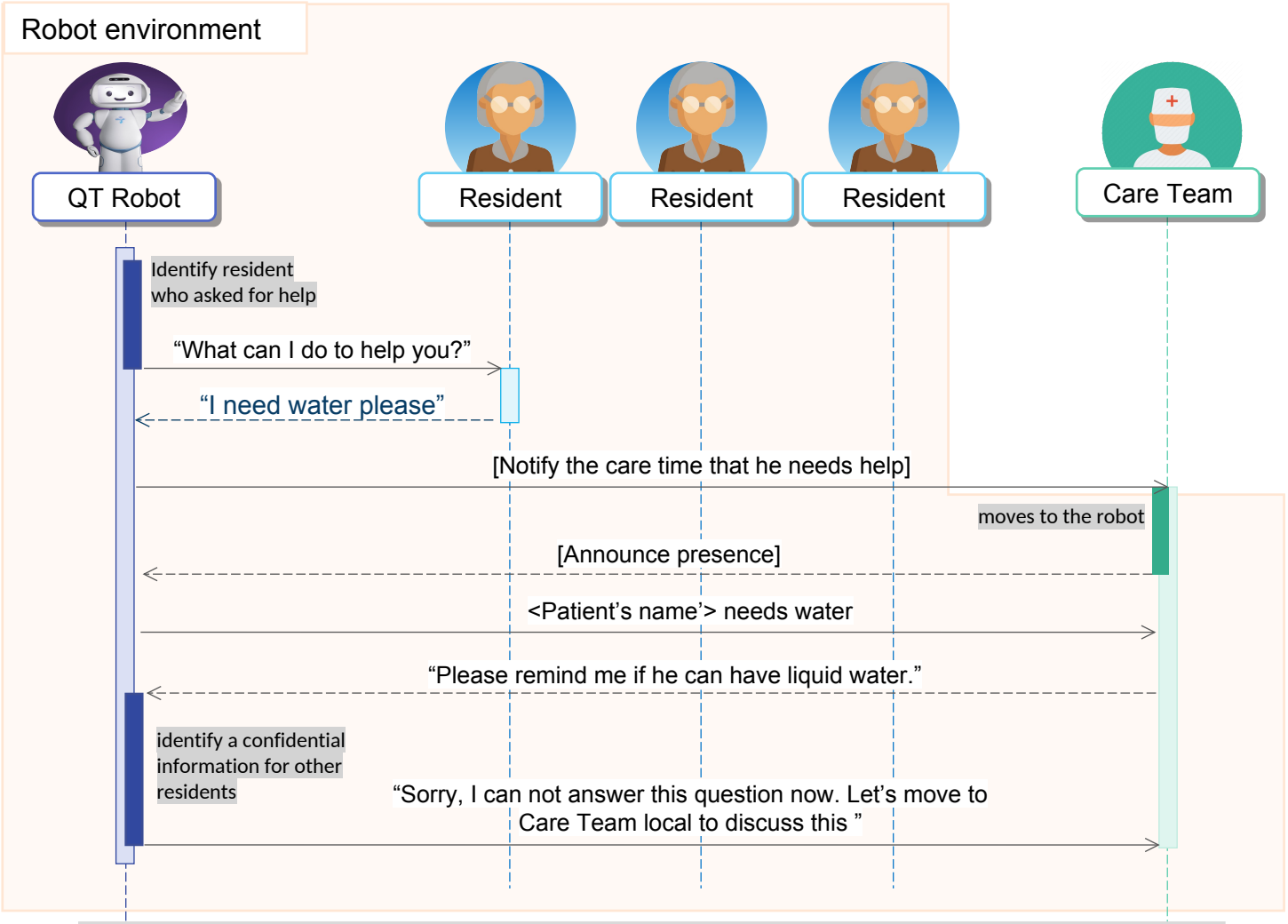
Context : The robot sees someone entering the room and starts an interaction



- Introducing to a new resident:
  - Creation of a new user profile in the knowledge base to remember the new resident
  - Asking the name for better identification (when reporting to the Care Team)
  - Proposing help, answer to questions...:
    - About diner (time, food...)
    - About appointments
    - About available activities
    - ...

One-to many interaction

Context : The robot is in a group of resident because one of them asked for help

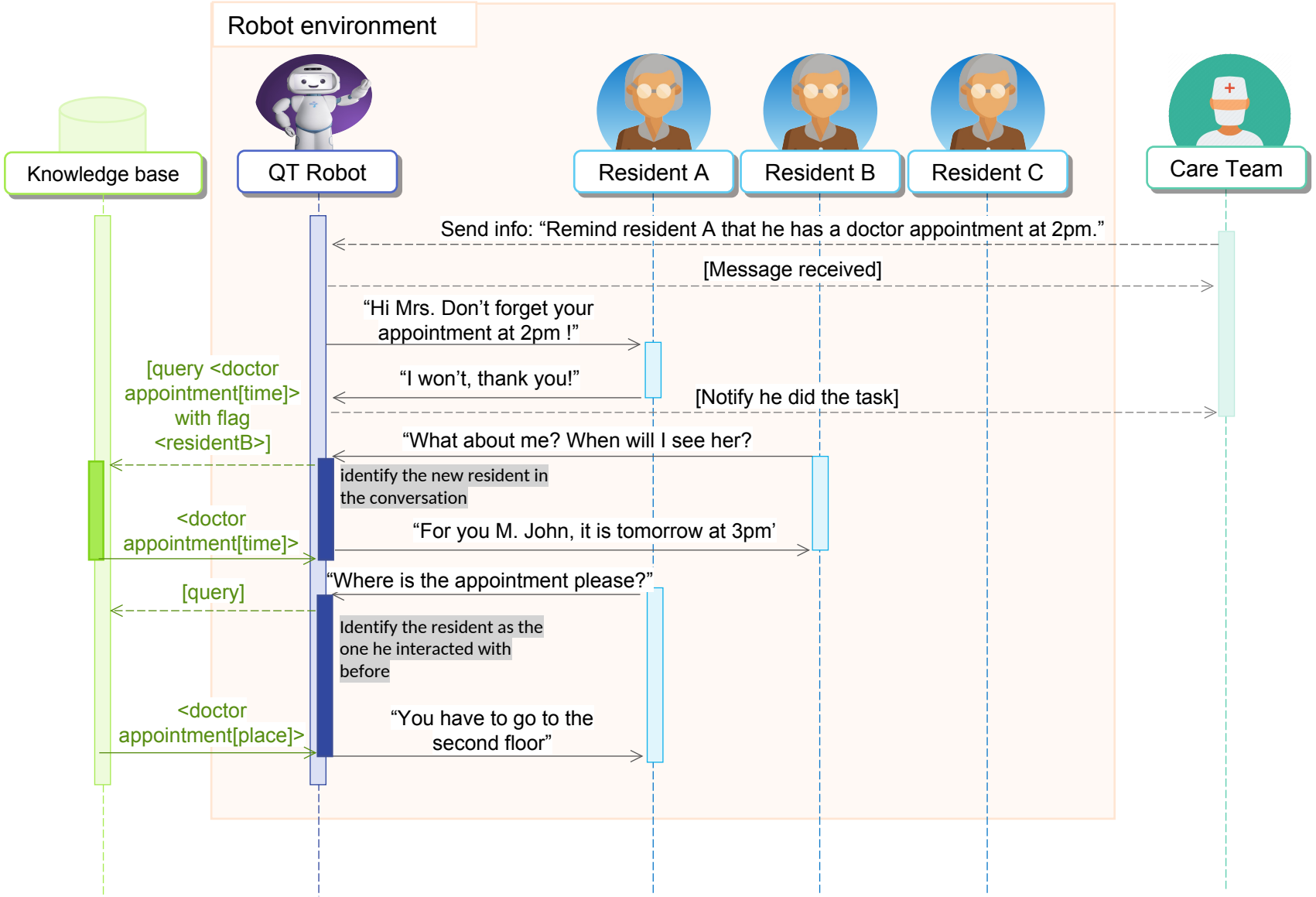


The robot and the Care Team Member move to a place without residents, where the question can be answered

- The robot knows confidential information that it can not say when unauthorized individuals are around
  - Needs to classify the information he receives (or is asked for), depending on the confidentiality.
  - Environment perception: the robot needs a good understanding of who is around, and who could hear what he is saying
- Some shared spaces could be considered as 'too risky' for sensitive information:

One-to many interaction

Context : The robot is in a group of resident because one of them asked for help



- The robot needs to be able to switch between different conversations
  - keep track of the recent conversations with the residents
  - Manage interruptions without forgetting someone
- The robot has to keep track of the tasks given be the care team
  - Different flags: success, fail, in progress
  - Go find help if he fails, notify the care team in case of success